ECLT5930B/SEEM5740B Engineering Economics and Cost Management

Chapter 11
Expected Utility Theory and Stochastic Dominance

Huifu Xu

Department of System Engineering and Engineering Management
The Chinese University of Hong Kong

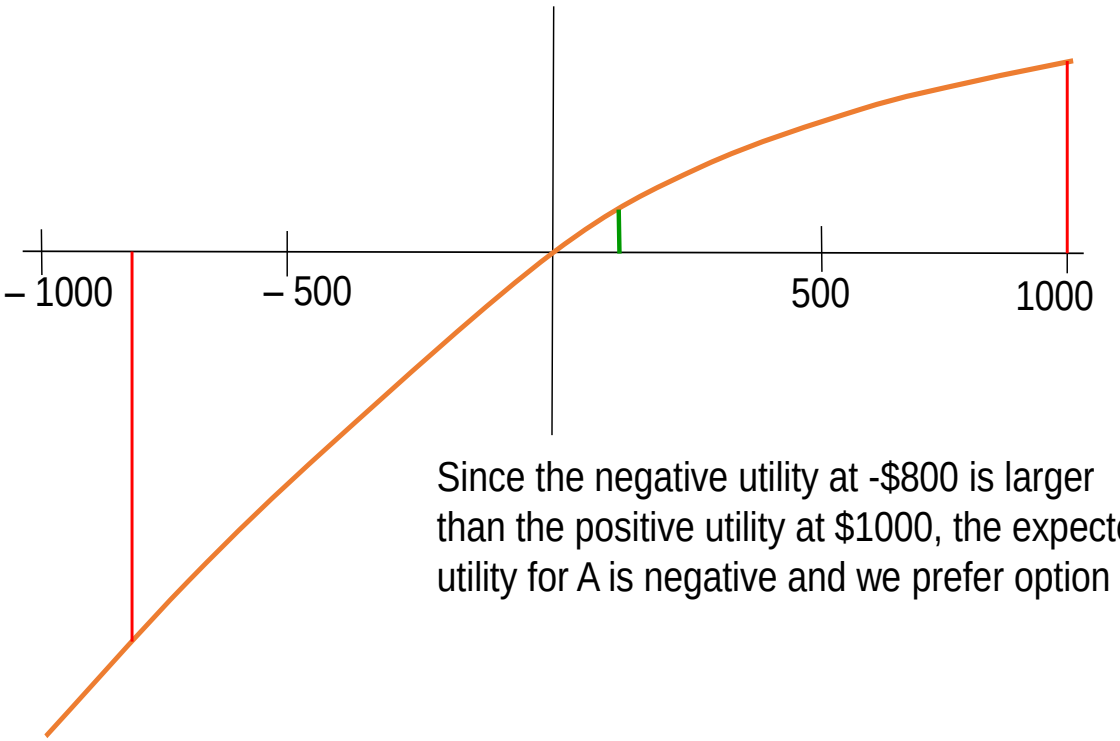
1. Motivation

A decision making problem

Example 10.1

- *Seven envelopes: one definitely contains $4,*
- *there is a $5 note in five of the other six and one envelope is empty.*
- *Will you take the $4 for sure or gamble on getting $5?*
- *Consider the worst? the best of the expected value?*

Comparing expected utility



Since the negative utility at -$800 is larger than the positive utility at $1000, the expected utility for A is negative and we prefer option B.

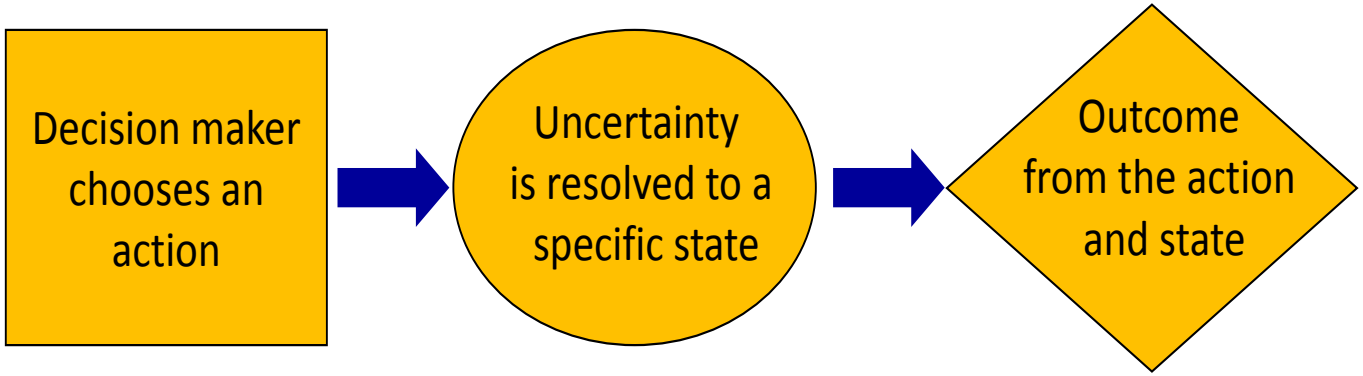
A	With prob 0.5, Gain $1000	With prob 0.5, Lose $800
B	With certainty, Gain $99	

Decision analysis is a formal way of determining optimal strategies when a decision maker is faced with several decision alternatives and an uncertain pattern of future events. There are essentially three components:

- Decision criteria
- Information on the uncertainties
- Finding an optimal decision from a set of feasible decisions

The focus will be on the case when there are finite number of feasible decisions.

Decisions, states & outcomes

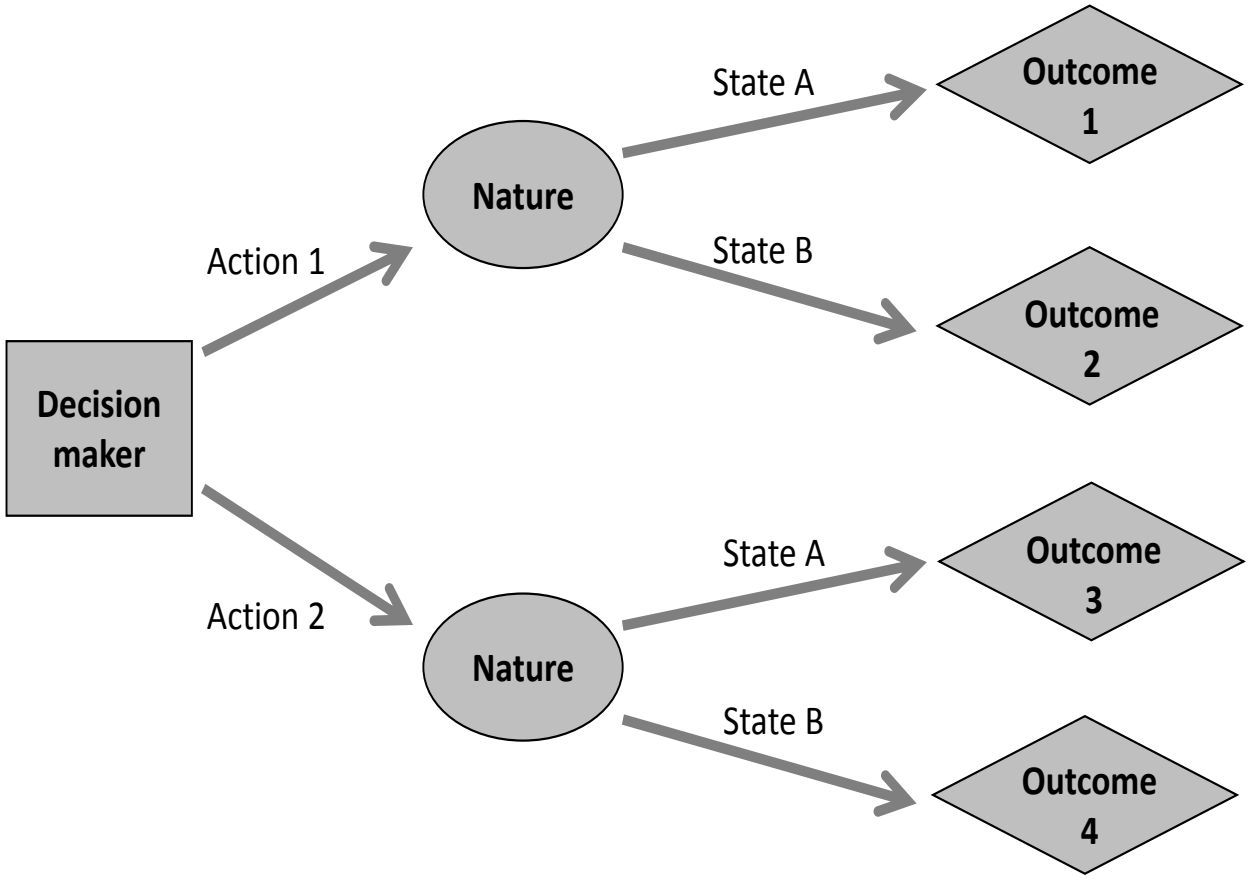


- The set of **actions** are our available decisions
- The **states** are all the things that might happen; or that have already happened but we don't know about
- A pair (action, state) implies a specific **outcome**
- Be careful: the word outcome is used in different ways by different authors

Probability, values and comparisons

- What are the probabilities associated with the various states that may occur? This is often a subjective judgement but we cannot easily avoid it.
- What value do the different outcomes deliver to us?
- Can we convert everything to dollars? How can we compare one possible decision leading to a particular set of outcomes (with associated probabilities and values), and another decision leading to a different set of outcomes?

A decision tree

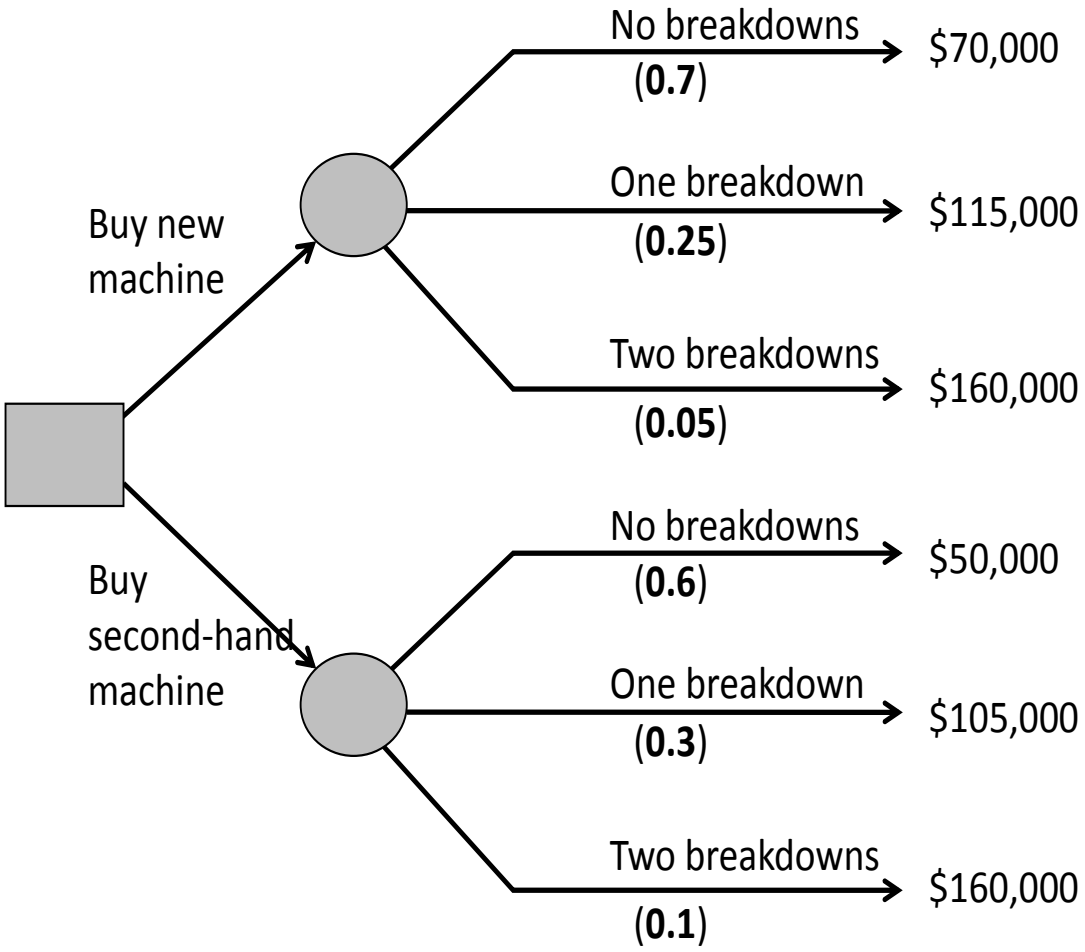


Matchstock example

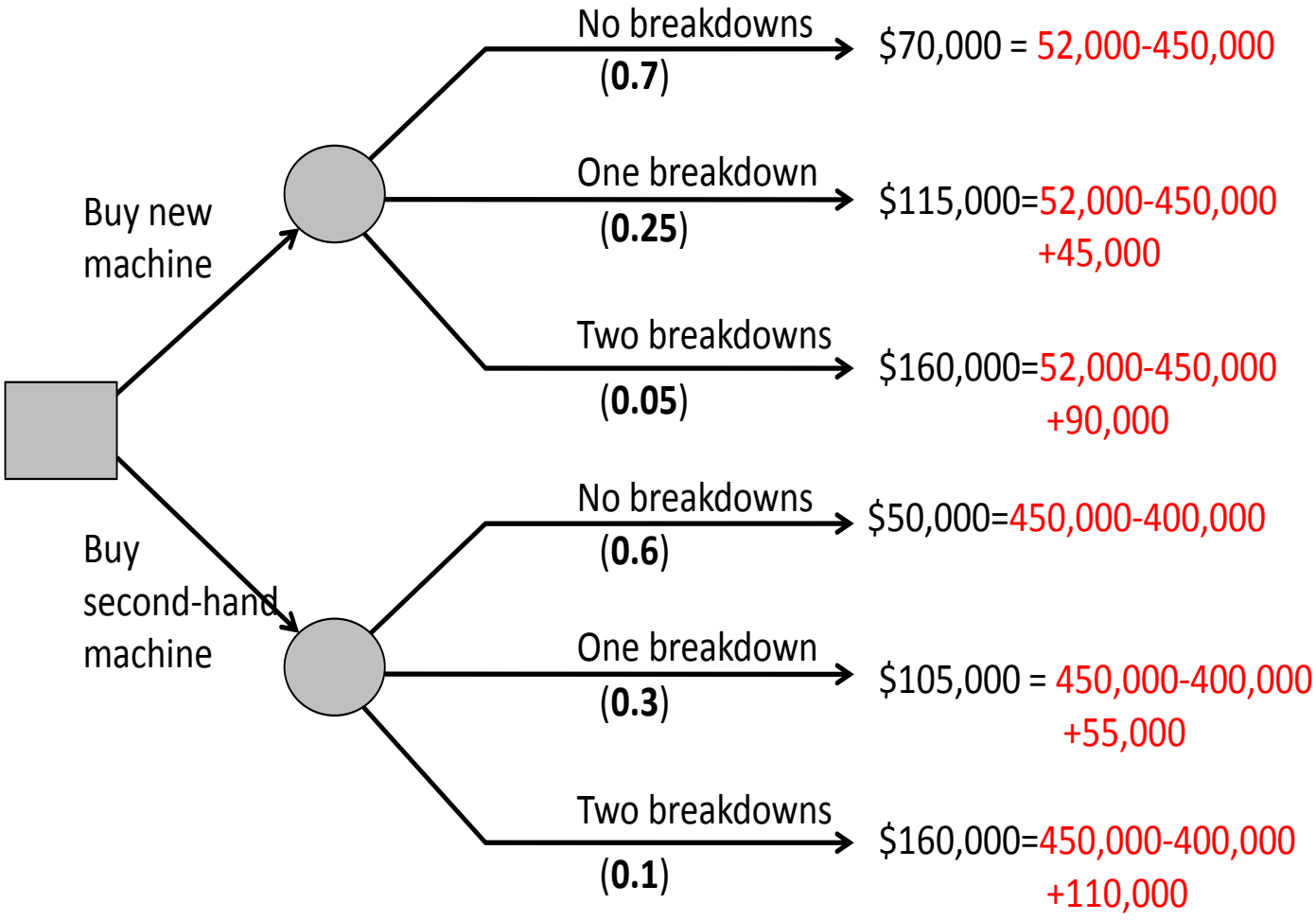
Example 10.3

- *Matchstock needs to buy a tunnelling machine. Cost (new) is* $\$520,000$. *It will be sold in* 2 *years for* $\$450,000$. *There is a* 0.25 *chance of a single breakdown and a* 0.05 *chance of* 2 *breakdowns during the two year project (with* 0.7 *probability of no breakdowns). Each breakdown costs* $\$45,000$.
- *A second hand machine costs* $\$450,000$, *and will be able to do the job satisfactorily. After two years it will be sold for* $\$400,000$. *There is a* 0.3 *chance of a single breakdown and a* 0.1 *chance of* 2 *breakdowns (with* 0.6 *probability of no breakdowns). Each breakdown costs* $\$55,000$ *in total.*

Matchstock decision tree



Matchstock decision tree



Buying a new tunnelling machine incurs expected costs (in \$1000s) of

$$0.7 \times 70 + 0.25 \times 115 + 0.05 \times 160 = 85.75$$

Buying a second hand machine gives expected costs of

$$0.6 \times 50 + 0.3 \times 105 + 0.1 \times 160 = 77.5.$$

So choosing the second hand tunnelling machine has lower costs and will maximize the expected profit.

St. Petersburg paradox

- A game is played where we are offered a prize that depends on coin tosses.
- If the first toss comes up heads we win $1 and the game finishes.
- If the first toss is a tail and the next toss is a head we win $2 and the game finishes.
- If the first two tosses are tails and the next is heads we win $4 and the game finishes, etc. Each time the amount we win doubles.
- How much would we pay to enter this game? Most people might pay $5 or may be $10 but no more. What is the expected profit?

St. Petersburg paradox

- How much would we pay to enter this game?
- Let X denote the win from the game. Then X is a random variable. The expected value of X is

$$\begin{aligned}
\mathbb{E}[X] &= \frac{1}{2} \times 1 + \frac{1}{2}\left(\frac{1}{2} \times 2 + \frac{1}{2}\left(\frac{1}{2} \times 4 + \frac{1}{2}(\cdots)\right)\right) \\
&= \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 4 + \frac{1}{16} \times 8 + \cdots \\
&= \infty.
\end{aligned}$$

- In practice, most people might wish to pay \$5 or \$10 but no more. What has gone wrong with the expected value theory?
- Very low probability events are neglected even the returns are significant!

St. Petersburg paradox

- The paradox is named after the journal's name *"the Commentaries of the Imperial Academy of Science of Saint Petersburg"* where Daniel Bernoulli published the problem and his solution in 1738.
- However, the paradox was invented by Nicolas (Daniel's cousin) Bernoulli who first stated it in a letter to Pierre Raymond de Montmort on September 9, 1713 (de Montmort 1713).

2. Von Neumann-Morgenstern Expected Utility Theory

Expected Utility Theory (EUT)

- Choice depends on our current financial circumstances and our taste for gambling.
- A better approach is to define an individual utility function that determines how valuable it is to him/her to end up with different amounts of money.
- This is a logical and appropriate way to proceed and we call it Expected Utility Theory.
- EUT has a long history. Cramer (1728) wrote to Bernoulli: The mathematicians estimate money in proportion to its quantity, and men of good sense in proportion to the usage that they may make of it.

Expected Utility Theory (EUT)

- Remarkably EUT can be derived from three (reasonable) axioms for choices: ordering; continuity and independence. Essentially there are no alternatives to EUT.
- Choices (or prospects) involve a whole set of possible outcomes each with a probability (like the choices A and B in the previous example).
- We can say a prospect X involves an outcome x_1 with probability p_1; an outcome x_2 with probability p_2; $\cdots$ and an outcome x_n with probability p_n .
- Mathematically X is a random variable with discrete distribution!

Axiom 1: Weak order

- Let $\mathcal{L}$ denote a prospect space (lotteries with finite support) and $X, Y, Z \in \mathcal{L}$.
- We want to use a binary relation $\succeq$ to describe DM's preferences.
- Write $X \succeq Y$ if X is preferred to Y.
- Completeness: for any $X, Y \in \mathcal{L}$, either $X \succeq Y$ or $Y \succeq X$ or both.
- Transitivity: for any $X, Y, Z \in \mathcal{L}$, if $X \succeq Y$ and $Y \succeq Z$, then $X \succeq Z$.
- A combination of completeness and transitivity is known as weak order.

Axiom 2: Continuity

- Continuity: for any X, Y, Z with $X \succeq Y$ and $Y \succeq Z$, there is a probability p such that

$$Y \succeq pX + (1-p)Z \text{ and } pX + (1-p)Z \succeq Y.$$

It means that we are indifferent between Y and the compound prospect $pX + (1-p)Z$.

Theorem 10.1 (Debreu (1964))

If a decision maker's preference satisfies <u>*weak order and continuity*</u>*, then there is a real valued choice function* $\Psi(\cdot)$ *mapping from the prospect space* $\mathcal{L}$ *to a real value such that*

$$X \succeq Y \iff \Psi(X) \geq \Psi(Y).$$

Axiom 3: Independence

- Independence: let $X, Y, Z \in \mathcal{L}$ with $X \succeq Y$.

$$pX + (1-p)Z \succeq pY + (1-p)Z$$

  for all $p \in (0, 1)$.
- It means that if we know that we choose X in preference to Y, we should still prefer X to Y even if there is some fixed chance of a different outcome Z.

von Neumann-Morgenstern expected utility theory

Theorem 10.2 (von Neumann and Morgenstern 1947)

Suppose that a preference relation $\succeq$ on the set of all prospects satisfies Axioms 1-3.

$$X \succeq Y \quad \Longleftrightarrow \quad \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)].$$

In this case the choice function takes the form $\Psi(Z) := \mathbb{E}[u(Z)]$.

Example 10.2 revisited

Choice A: With probability 0.5 gain \$1000 and with probability 0.5 lose \$800;

Choice B: With certainty gain \$99.

$$u(99) > 0.5u(-800) + 0.5u(1000)$$

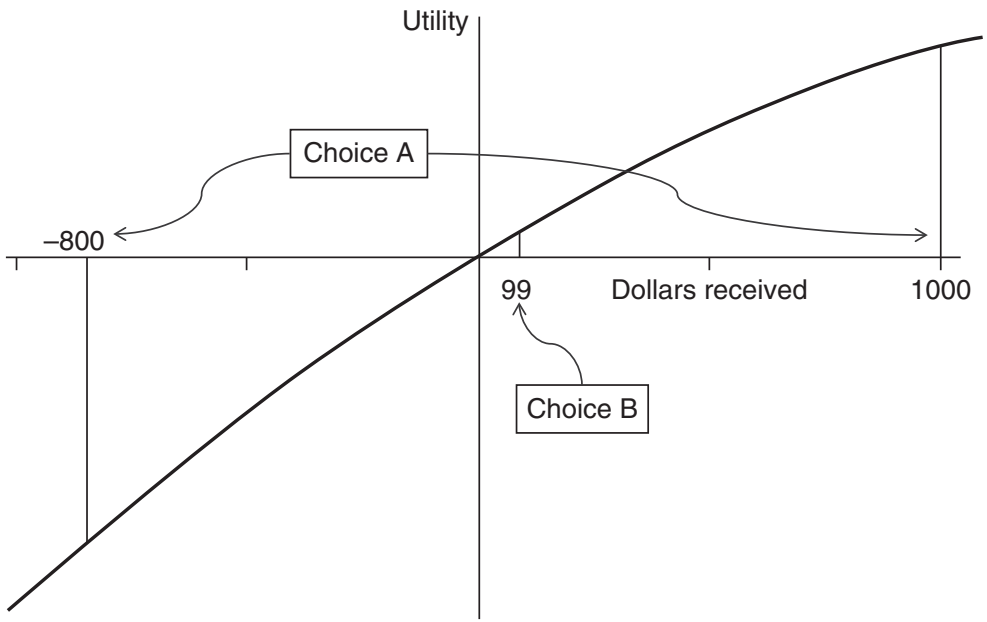


Figure: Characterizing preference with a utility function

Decision maker's attitude to risk: characterization from EUT

- A person is given the choice between two scenarios, one with a guaranteed payoff and one without. In the guaranteed scenario, the person receives $50.
- In the uncertain scenario, a coin is flipped to decide whether the person receives $100 or nothing.
- The expected payoff for both scenarios is $50, meaning that an individual who was insensitive to risk would not care whether they took the guaranteed payment or the gamble.

However, individuals may have different risk attitudes.

- A person is said to be: risk-averse (or risk-avoiding) - if he or she would accept a certain payment (certainty equivalent) of less than \$50 (for example, \$40), rather than taking the gamble and possibly receiving nothing.
- risk-neutral - if he or she is indifferent between the bet and a certain \$50 payment.
- risk-loving (or risk-seeking) - if he or she would accept the bet even when the guaranteed payment is more than \$50 (for example, \$60).

Risk aversion – risk aversion

Let X denote the random return and $u(\cdot)$ be the decision maker's utility function.

- The decision maker prefers to taking $\mathbb{E}[X]$ for certain gambling on X means

$$\Psi(X) := \mathbb{E}[u(X)] \leq u(\mathbb{E}[X]) \tag{1}$$

- Mathematically (2) holds if and only if u is a concave function. Example $u(x) = \ln x$.

An investment example

Example 10.4

- *There are 2 investment funds: growth and stable.*
- *In* **bad years** *the stable fund loses* 5% *and the growth fund loses* 25%.
- *In* **good years** *the stable fund gains* 15% *and the growth fund gains* 30%.
- *In* **medium years** *the stable fund gains* 5% *and the growth fund gains* 10%.
- *An investor estimates that bad years occur three years in ten, good years occur three years in ten, and medium years occur four years in ten.*
- *The investor's utility for* x *thousand dollars is given by* $\ln(x)$. *Given* $\$100,000$ *to invest, which option has the highest expected utility?*

	Bad year	Medium year	Good year	Expected value
Probability	0.3	0.4	0.3	
Stable fund:				
Value	95	105	115	105
Utility	ln(95) = 4.554	ln(105) = 4.654	ln(115) = 4.745	4.651
Growth fund:				
Value	75	110	130	105.5
Utility	ln(75) = 4.317	ln(110) = 4.701	ln(130) = 4.867	4.636

Certainty equivalent

Let Z be a lottery and u be a utility function. Define

$$C_u(Z) := u^{-1}(\mathbb{E}[u(Z)]),$$

which is the sure amount for which a decision maker remains indifferent to the lottery.

Let X and Y denote respectively the stable fund and growth fund. Then

- $C_u(X) := u^{-1}(\mathbb{E}[u(X)]) = e^{4.651} = 104.69.$
- $C_u(Y) := u^{-1}(\mathbb{E}[u(Y)]) = e^{4.636} = 103.13.$
- They are both less than the expected value 105 because the utility function is concave!

Risk aversion – risk aversion

Let X denote the random return and $u(\cdot)$ be the decision maker's utility function.

- The decision maker prefers to taking $\mathbb{E}[X]$ for certain gambling on X means

$$\Psi(X) := \mathbb{E}[u(X)] \leq u(\mathbb{E}[X]) \tag{2}$$

- Mathematically (2) holds if and only if u is a concave function. Example $u(x) = \ln x$.

Concave functions

Let $f(x)$ be a univariate function. If for any $x_1, x_2 \in \mathbb{R}$

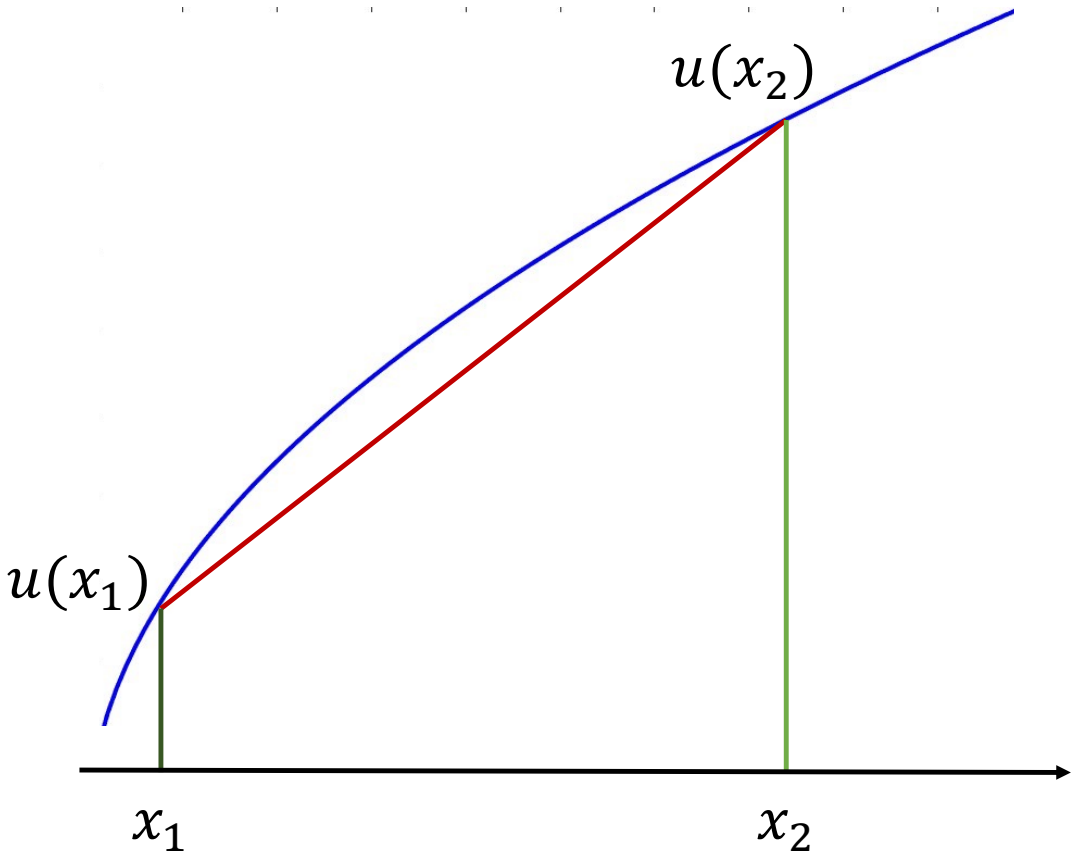
$$f(tx_1 + (1-t)x_2) \geq tf(x_1) + (1-t)f(x_2), \ \forall t \in [0,1],$$

then f is said to be *concave* on $\mathbb{R}$.
When f is twice continuously differentiable, a *sufficient* condition for f to be convex is

$$f''(x) \leq 0, \forall x \in \mathbb{R}.$$

Geometric interpretation?

Geometric interpretation of concavity



Risk seeking – convex utility function

- The decision maker prefers gambling to taking $\mathbb{E}[X]$ for certain means

$$U(X) := \mathbb{E}[u(X)] \geq u(\mathbb{E}[X]) \tag{3}$$

- Mathematically (3) holds if and only if u is a convex function.

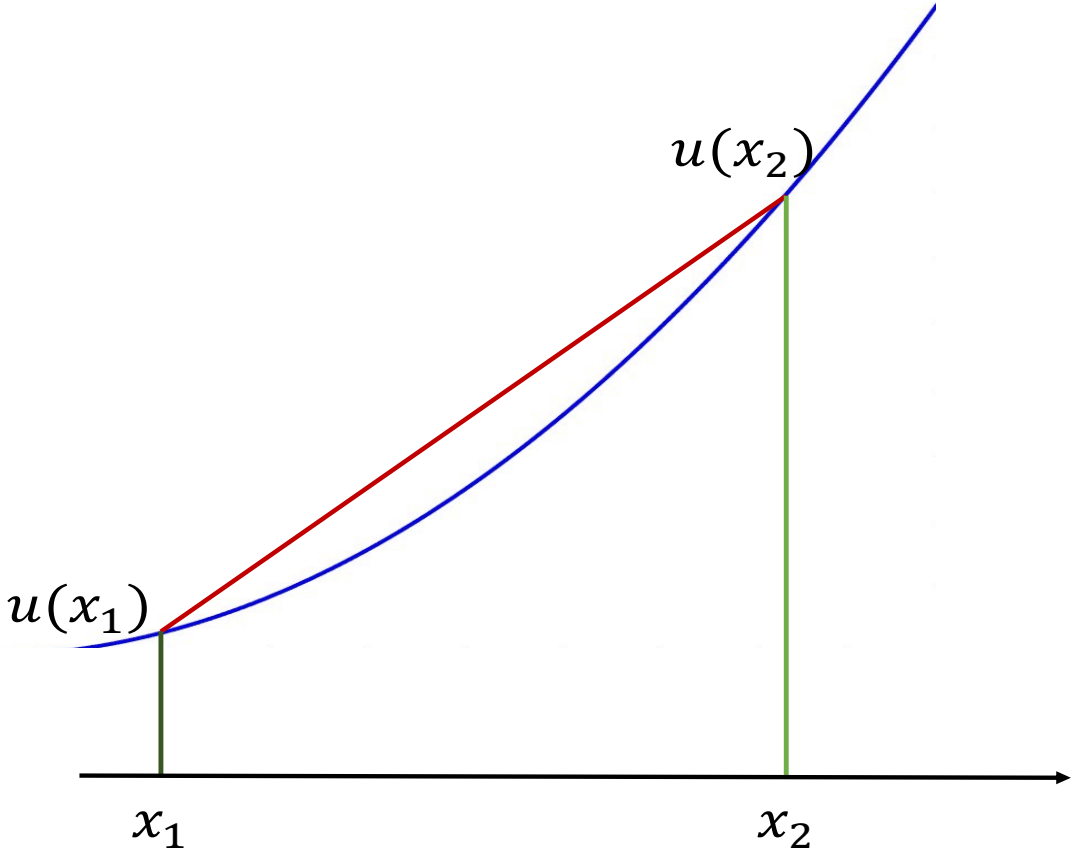
Convex functions

Let $f(x)$ be a univariate function. If for any $x_1, x_2 \in \mathbb{R}$

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2), \ \forall t \in [0,1],$$

then f is said to be *convex* on $\mathbb{R}$.
Geometric interpretation?

Geometric interpretation of convexity



Risk neutral seeking

- The decision maker is indifferent between gambling and taking $\mathbb{E}[X]$ for certain means

$$U(X) := \mathbb{E}[u(X)] = u(\mathbb{E}[X]) \tag{4}$$

- Mathematically (4) holds if and only if u is a linear function.

Example 10.5

James has \$1000 *which he wants to invest for a year. He can put the money into a savings account, which pays an interest rate of* 4%. *His friend Kate asks him to invest the money in a business venture for which she needs exactly* \$1000. *However, Kate's business will fail with probability* 0.3 *and if this happens James will get nothing. On the other hand, if the business succeeds it will make* \$2000 *and this money can be used to repay the loan to James. The time taken to repay the money if the venture succeeds will be one year.*

Using EUT and assuming that James is risk neutral, how much would Kate have to repay James in order to convince James to lend her the money?

Answers:

From saving, James is guaranteed to obtain $\$1000 \times 4\% = \$40$ profit.

Let x denote the minimum amount that Kate should pay James on the top of the loan. James expected return will be

$$0.3 \times 0 + 0.70 \times (1000 + x).$$

So

$$0.3 \times 0 + 0.7 \times (1000 + x) \geq 1040$$

$x \geq 485$. This means Kate should repay James at least 1485 to convince James to lend to her his money.

Assume James is risk averse with concave utility function $u(x) = \sqrt{x}$. How much would Kate have to repay James if the business venture succeeds in order to convince James to lend her the money?

Answers: If James uses the utility function $u(x) = \sqrt{x}$, then

$$0.3 \times \sqrt{0} + 0.7 \times \sqrt{1000 + x} \geq \sqrt{1040}$$

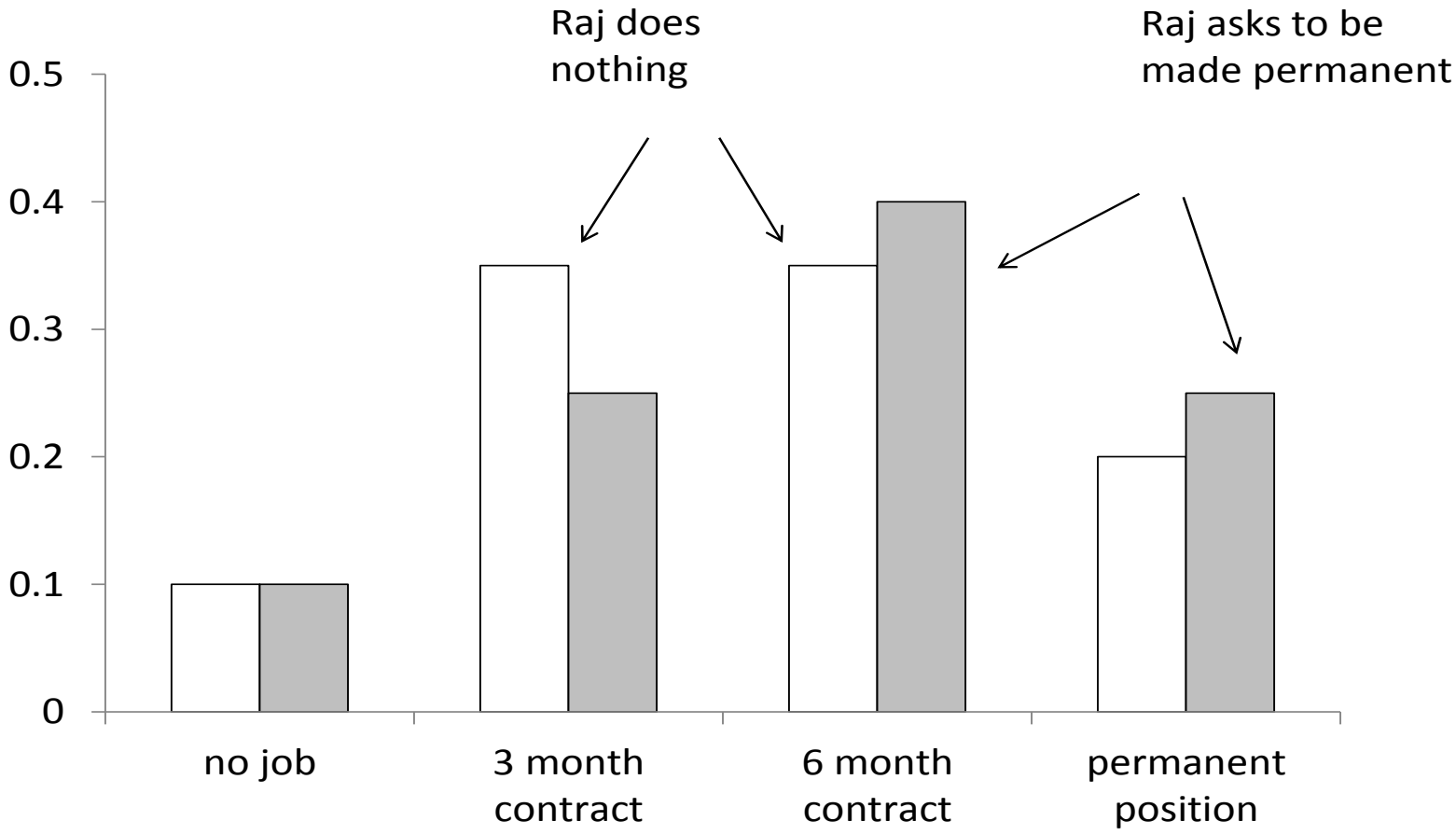
Solving the inequality above, we obtain $x \geq 1122$ which means Kate should pay James at least \$2122 in order to convince him lending \$1000 to her.

3. Stochastic Dominance

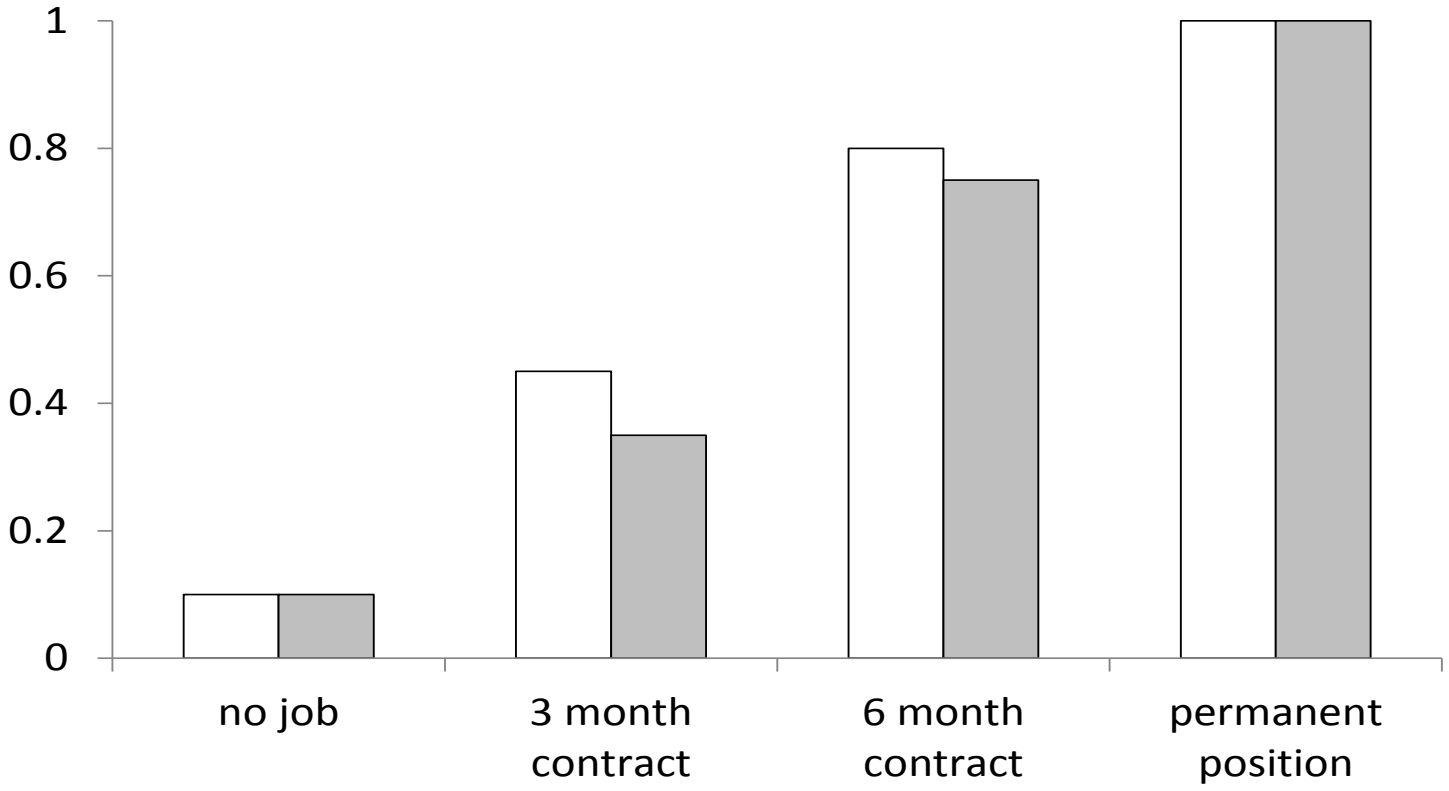
Stochastic dominance

- Raj has a contract position for 6 months. He thinks there is a 10% chance of his job not being renewed at the end of the contract, a 35% chance of it being renewed for 3 months, a 35% chance of it being renewed for 6 months and a 20% chance of the job being made permanent.
- Raj is considering going to his boss and asking to be made permanent. He thinks this will increase the chance of his being made permanent to 25% with probabilities of the other outcomes being: 10% job not renewed; 25% job renewed for 3 months; 40% job renewed for 6 months. What should he do?

A risk profile



A cumulative risk profile



First order stochastic dominance

Let X and Y be two random variables. Let $F_X(t)$ and $F_Y(t)$ denote their cumulative distribution functions, i.e.,

$$F_X(t) = P(X \leq t)$$

and

$$F_Y(t) = P(Y \leq t).$$

X is said to dominate Y in first order if

$$F_X(t) \leq F_Y(t), \forall t \tag{5}$$

In the case when the cumulative distribution functions are monotonically increasing, it means

$$F_X^{-1}(\beta) \geq F_Y^{-1}(\beta), \forall \beta \in [0, 1]. \tag{6}$$

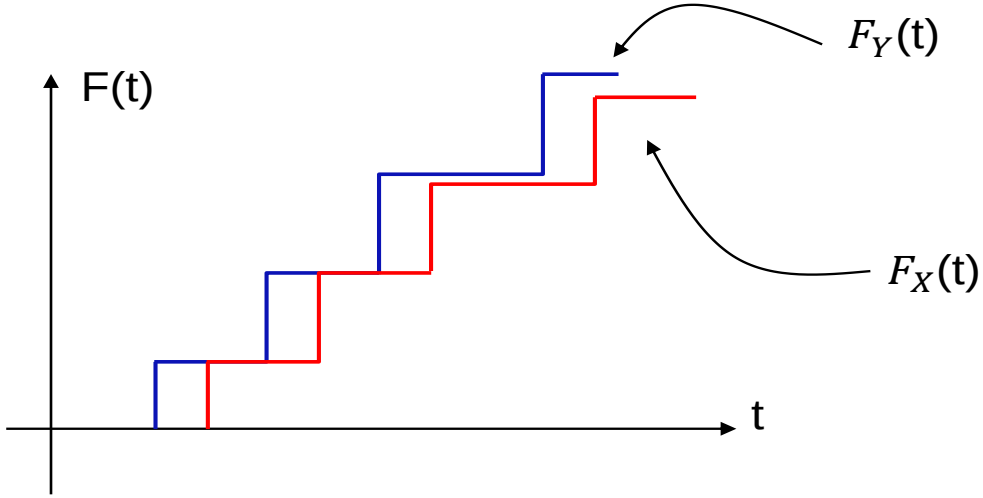
Second order stochastic dominance

X is said to dominate Y in second order if

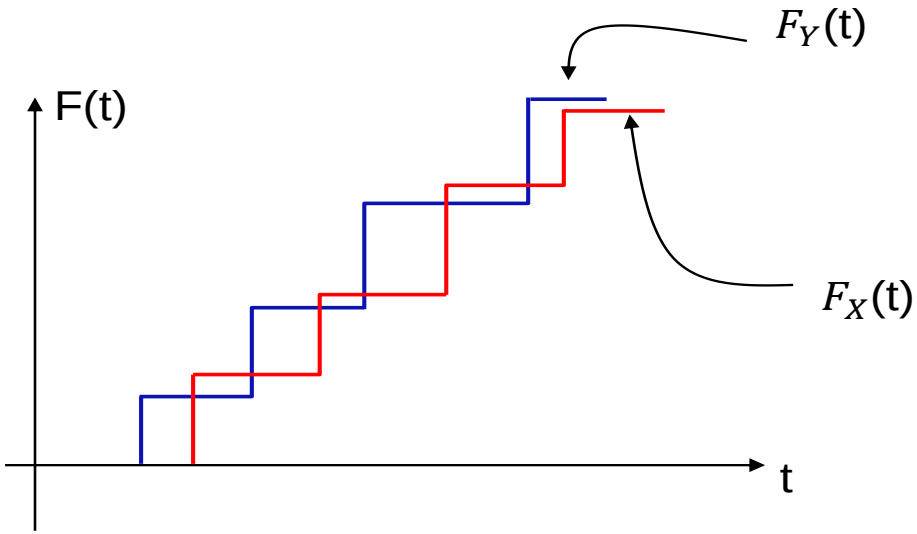
$$\int_{\infty}^{t} F_X(\eta)d\eta \leq \int_{\infty}^{t} F_Y(\eta)d\eta. \tag{7}$$

First order dominance implies second order dominance!

1st order dominance

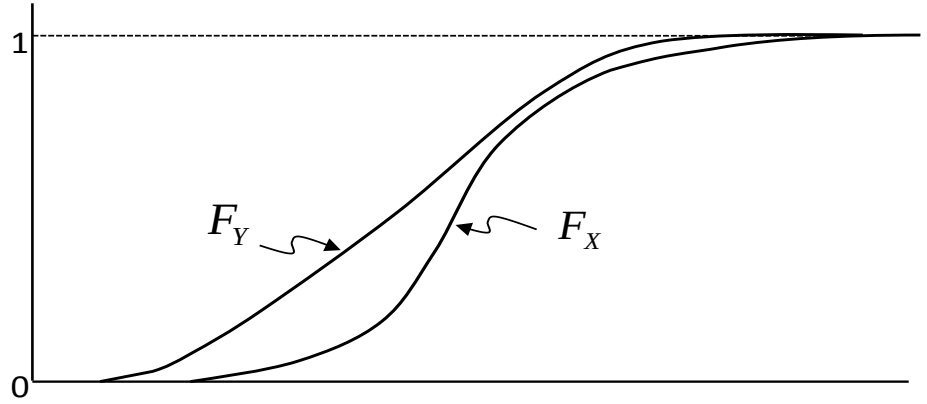


2nd order dominance

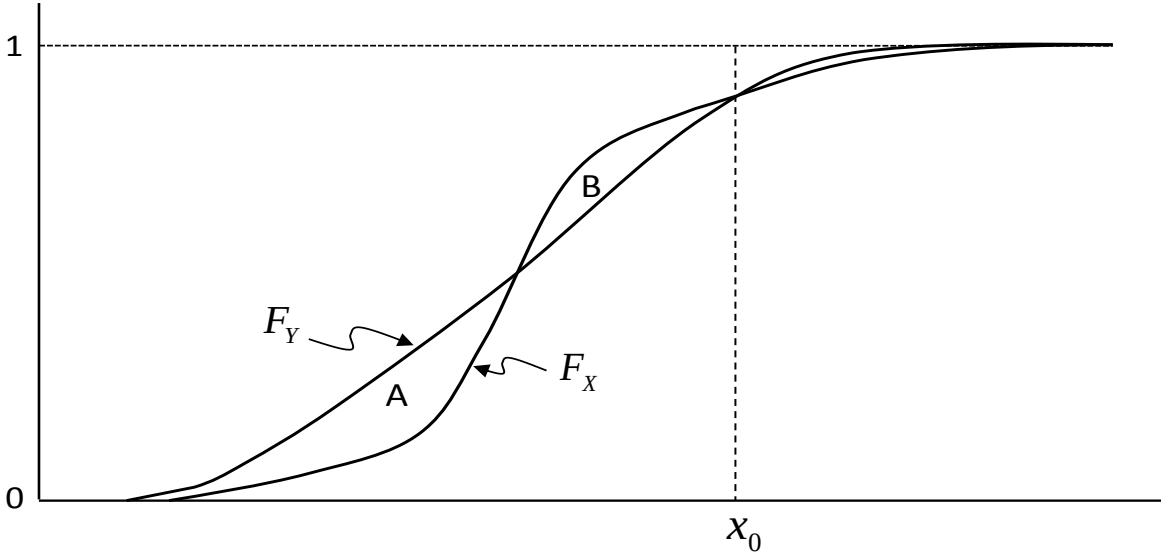


Stochastic dominance and CDFs

- First order stochastic dominance:
  X dominates Y if the CDF for X is **lower**



- Second order stochastic dominance (region A larger than region B



Example of first order dominance

- The prospect $A = (\$100, 0.2;\ \ \$160, 0.2; \$200, 0.6)$ stochastically dominates the prospect $B = (\$100, 0.3; \$120, 0.1; \$180, 0.1; \$200, 0.5)$

amount	A	c.d.f sum for A	B	c.d.f sum for B
$100	0.2	0.2	0.3	0.3
$120	0	0.2	0.1	0.4
$160	0.2	0.4	0	0.4
$180	0	0.4	0.1	0.5
$200	0.6	1	0.5	1

Example 10.6

Let X be a random variable which follows exponential distribution with density function $\rho(x) = 0.1e^{-0.1x}$ where $x \geq 0$. Let M be a positive number and Y be a discrete random variable

$$Y = \begin{cases} 0 & \text{with probability } 1 - e^{-0.1M}, \\ M & \text{with probability } e^{-0.1M}. \end{cases}$$

Show that (a) X dominates Y in first order and (b) $\mathrm{CVaR}_\beta(X) \geq \mathrm{CVaR}_\beta(Y)$ for any $\beta \in (0, 1)$.

Proof.

Part (a). We calculate the c.d.f of X and Y respectively.

$$F_X(x) = \begin{cases} 1 - e^{-0.1x}, \text{for } x \in [0, M), \\ 1 - e^{-0.1x}, \text{for } x \in [M, \infty). \end{cases}$$

$$F_Y(x) = \begin{cases} 1 - e^{-0.1M}, \text{for } x \in [0, M), \\ 1, \text{for } x \in [M, \infty). \end{cases}$$

It is easy to observe that

$$F_X(x) \leq F_Y(x), \forall x \in [0, \infty).$$

Part (b). It follows by the definition of VaR.

Example of second order dominance first order dominance failed

- The prospect $A = (\$100, 0.2;\ \ \$160, 0.2; \$200, 0.6)$ stochastically dominates the prospect

$B = (\$100, 0.3; \$120, 0.1; \$180, 0.1; \$200, 0.5)$

amount	A	c.d.f of A	B	c.d.f of B
$100	0.2	0.2	0.3	0.3
$120	0.1	0.3	0.1	0.4
$160	0.2	0.5	0	0.4
$180	0	0.5	0.1	0.5
$200	0.5	1	0.5	1

Example of second order dominance second order dominance satisfied

- The prospect $A = (\$100, 0.2;\ \ \$160, 0.2; \$200, 0.6)$ stochastically dominates the prospect

  $B = (\$100, 0.3; \$120, 0.1; \$180, 0.1; \$200, 0.5)$

amount	A	Integral of c.d.f A	B	Integral of c.d.f B
$100	0.2	0	0.3	0
$120	0.1	4	0.1	6
$160	0.2	16	0	22
$180	0	26	0.1	30
$200	0.5	36	0.5	40

1st order dominance vs EUT

Theorem 10.3 (Hanoch and Levy (1969))

Let X and Y be two random variables with c.d.f. F and G respectively. X first-order stochastically dominates Y if and only if

$$\mathbb{E}[u(X)] = \int u(x)dF(x) \geq \int u(y)dG(y) = \mathbb{E}[u(Y)]$$

for any increasing utility function $u(\cdot)$.

2nd order dominance vs EUT

Theorem 10.4 (Hadar and Russell (1969))

Let X and Y be two random variables with c.d.f. F and G respectively. X second-order stochastically dominates Y if and only if

$$\mathbb{E}[u(X)] = \int u(x)dF(x) \geq \int u(y)dG(y) = \mathbb{E}[u(Y)]$$

for any increasing concave utility function $u(\cdot)$.

4. Beyond VNM's Expected Utility Theory

Allais paradox

Par M. Allias [1] found something contradictory to von Neumann and Morgenstern theorem.

Example 10.7

- DECISION 1: Choose between
  A: $\$2,500$ with probability 0.33, $\$2,400$ with probability 0.66, 0 with probability 0.01;
  B: $\$2,400$ with certainty.
  More than 80% of people choose B
- DECISION 2: Choose between
  C: $\$2,500$ with probability 0.33, 0 with probability 0.67
  D: $\$2,400$ with probability 0.34, 0 with probability 0.66.
  More than 80% of people choose C

Implications to EUT

- From Decision 1 we have (since $u(0) = 0$)

$$u(2400) > 0.33u(2500) + 0.66u(2400)$$

which implies

$$0.34u(2400) > 0.33u(2500)$$

But from Decision 2 we have the opposite:

$$0.34u(2400) < 0.33u(2500)$$

Decision 2 is obtained by eliminating a 0.66 chance of winning \$2400 from both prospects in Decision 1. This gives more reduction in desirability when it turns a sure gain to a probable one, rather than when it simple decreases the chance of a gain

Preference for sure gain

Example 10.8

- *DECISION 1: Choose between*
  A: \$4000 *with probability* 0.8, \$0 *with probability* 0.2
  B: \$3000 *with certainty*
- *DECISION 2: Choose between*
  C: \$4000 *with probability* 0.2, \$0 *with probability* 0.8
  D: \$3000 *with probability* 0.25, \$0 *with probability* 0.75

80% *of people choose B in problem 1 and* 65% *of people choose C in problem 2*

Implication to EUT

- From Decision 1 we can deduce that

$$u(3000) > 0.8u(4000)$$

From Decision 2 we can deduce that

$$0.2u(4000) > 0.25u(3000)$$

which is equivalent to

$$u(3000) < 0.8u(4000)$$

The choice of B in Decision 1 reflects risk aversion, and a curvature of the utility downwards. This should be enough to imply that D is preferred to C in Decision 2, but it turns out not to be the case for a majority.

Violation of independence axiomatic property

Independence: let $X, Y, Z \in \mathcal{L}$ with $X \succeq Y$.

$$pX + (1-p)Z \succeq pY + (1-p)Z$$

for all $p \in (0,1)$.

A		B	
2400	66%	2400	66%
2500	33%	2400	34%
0	1%		

C		D	
0	66%	0	66%
2500	33%	2400	34%
0	1%		

Violation of independence axiomatic property

Independence: let $X, Y, Z \in \mathcal{L}$ with $X \succeq Y$.

$$pX + (1-p)Z \succeq pY + (1-p)Z$$

for all $p \in (0,1)$.

Y		X	
2500	33/34	2400	34/34
0	1/34		

Let $Z = 2400$ and $p = 0.34$.

A=0.34*Y+0.66Z		B=0.34*X+0.66Z	
2400	66%	2400	66%
2500	33%	2400	34%
0	1%		

Violation of independence axiomatic property

Independence: let $X, Y, Z \in \mathcal{L}$ with $X \succeq Y$.

$$pX + (1-p)Z \succeq pY + (1-p)Z$$

for all $p \in (0,1)$.

Y		X	
2500	33/34	2400	34/34
0	1/34		

Let $Z = 0$ and $p = 0.34$.

C=0.34*Y+0.66Z		D=0.34*X+0.66Z	
0	66%	0	66%
2500	33%	2400	34%
0	1%		

Loses rather than gains

Example 10.9

The opposite results are obtained when losses are involved

- *PROBLEM 1: Choose between*
  A: loss of \$4000 *with prob. 0.8,* \$0 *with prob.* 0.2
  B: loss of \$3000 *with certainty*
- *PROBLEM 2: Choose between*
  C: loss of \$4000 *with prob.* 0.2, \$0 *with prob.* 0.8
  D: loss of \$3000 *with prob.* 0.25, \$0 *with prob.* 0.75

90% *of people choose A in problem 1, but* 58% *of people choose D in problem 2*

In summary

- In general people are risk averse over positive gains (preferring not to gamble).
- People are risk seeking over negative gains (i.e. losses). They will gamble to give themselves a chance of avoiding a loss.
- There is a divergence from EUT that in these examples seems to be associated with an excess preference for a sure gain (and an excess dislike of a sure loss).

5. Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR)

Value-at-Risk (VaR): motivation

- Consider a random variable L which represents loss. Let $F_L(x)$ denote the cumulative distribution function (cdf) of L, i.e.,

$$F_L(x) = P(L \leq x).$$

- The cdf tells us the probability of loss less or equal to x is $F_L(x)$.
- Equivalently we can say with probability $1 - F_L(x)$ the loss exceeds level x (we are effectively attaching the probability to a consequence), i.e.,

$$P(L > x) = 1 - P(L \leq x) = 1 - F_L(x) =: G_L(x).$$

Value-at-Risk (VaR): motivation

- In practice, we often say with 95% confidence, the loss falls below \$1 million.
- If we specify the confidence level first, then how do we figure out the corresponding level of loss?

Value-at-Risk (VaR)

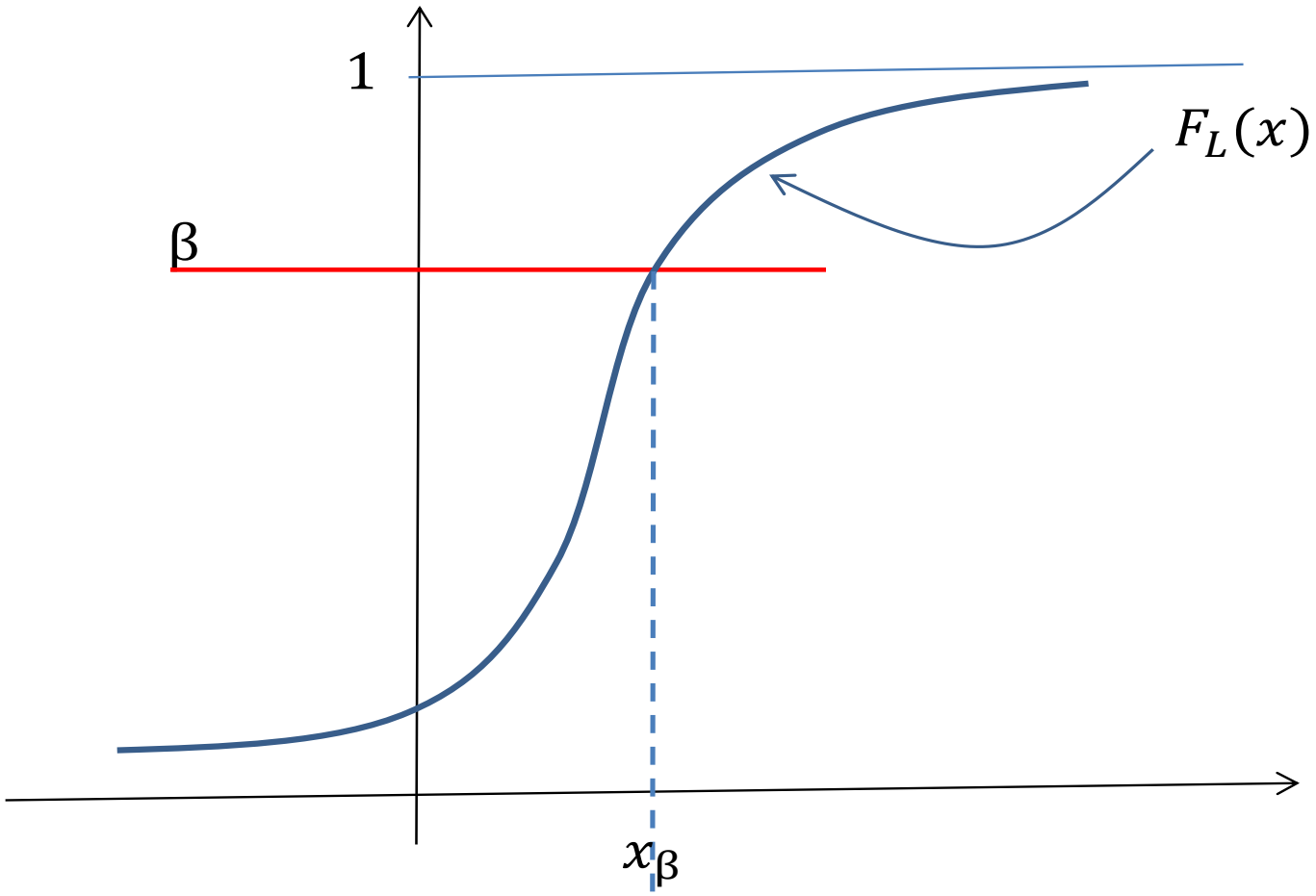
- Mathematically, we are given a value β (usually close to 1) and then we are asked to identify x such that

$$F_L(x) := P(L \leq x) = \beta \tag{8}$$

or equivalently

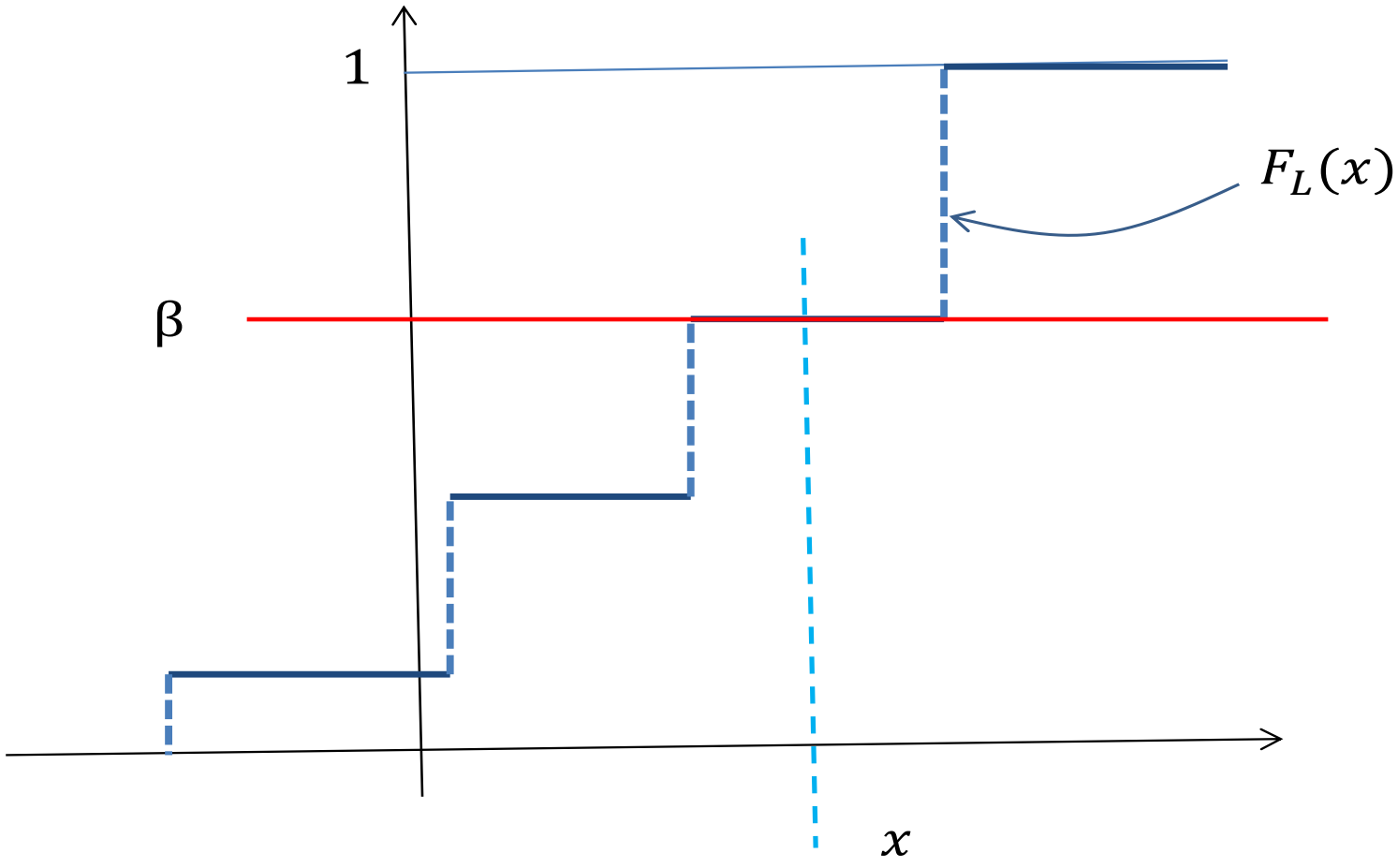
$$1 - F_L(x) := P(L > x) = 1 - \beta. \tag{9}$$

- If $F_L(\cdot)$ is continuous and monotonically increasing, then we can solve from (8) a unique solution $x_\beta = F_L^{-1}(\beta)$, where F_L^{-1} denotes the inverse of F_L.
- We call x_β Value-at-Risk.



Unique x_β corresponding each β value

- When L follows a discrete distribution, $F_L(x)$ is a step function which is not strictly monotonically increasing and it is discontinuous at some points. In that case, (8) and (9) may have multiple solutions or no solution.



Discrete distribution: multiple x corresponding some β values

Value-at-Risk (VaR) – formal definition

- A strict mathematical definition of VaR is

$$\mathrm{VaR}_\beta(L) = \inf\{x : P(L \leq x) \geq \beta\} \tag{10}$$

or equivalently

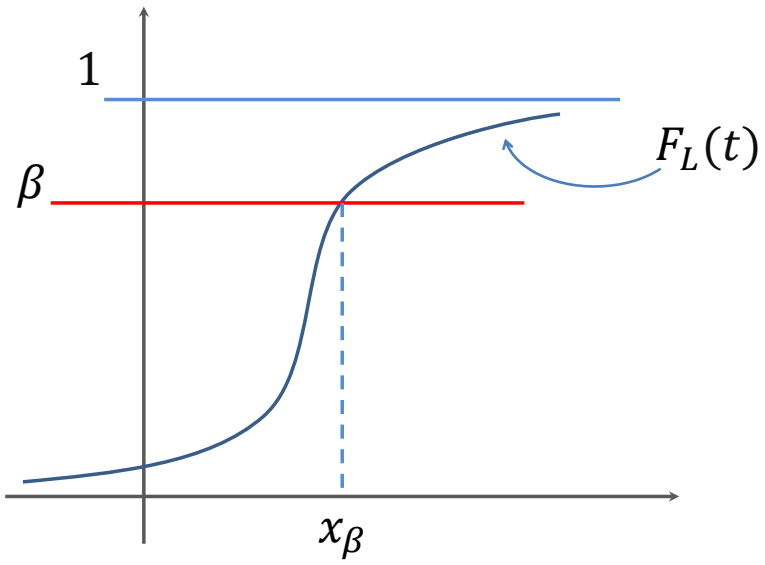
$$\mathrm{VaR}_\beta(L) = \inf\{x : P(L > x) < 1 - \beta\} \tag{11}$$

From (10), we can see that VaR_β is the first point of x at which $F_L(x)$ exceeds the specified level β and from (11) it may be interpreted as the minimum cash to be injected to a financial position such that the probability of loss after the injection being positive is less than $1 - \beta$.

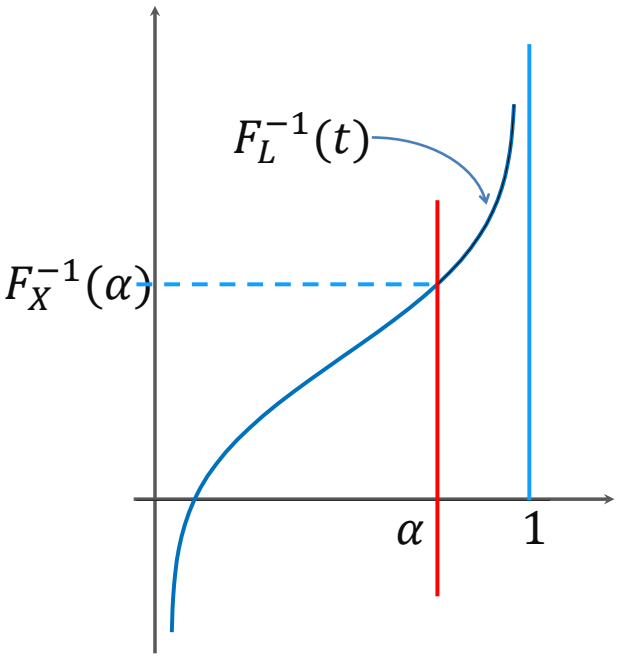
- For fixed L, $\mathrm{VaR}_\beta(L)$ may be viewed as a function of β, in that case, we write

$$F_L^{\leftarrow}(\beta) := \mathrm{VaR}_\beta(L) = \inf\{x : F_L(x) \geq \beta\}. \tag{12}$$

$F_L^{\leftarrow}(\beta)$ is known as quantile function.



Unique x_β corresponding each β value



Quantile function of L

Example 10.10

Let $L(\xi)$ denote the loss with the following probability distribution

$$P(L = 0.5) = 0.2,\ P(L = 1) = 0.3,$$
$$P(L = 1.5) = 0.2,\ P(L = 3) = 0.3.$$

Work out $\mathrm{VaR}_{0.95}$, $\mathrm{VaR}_{0.7}$ and $\mathrm{VaR}_{0.65}$.

By definition

$$\mathrm{VaR}_{0.95}(L) = \inf\{x : P(L \leq x) \geq 0.95\} = 3.$$

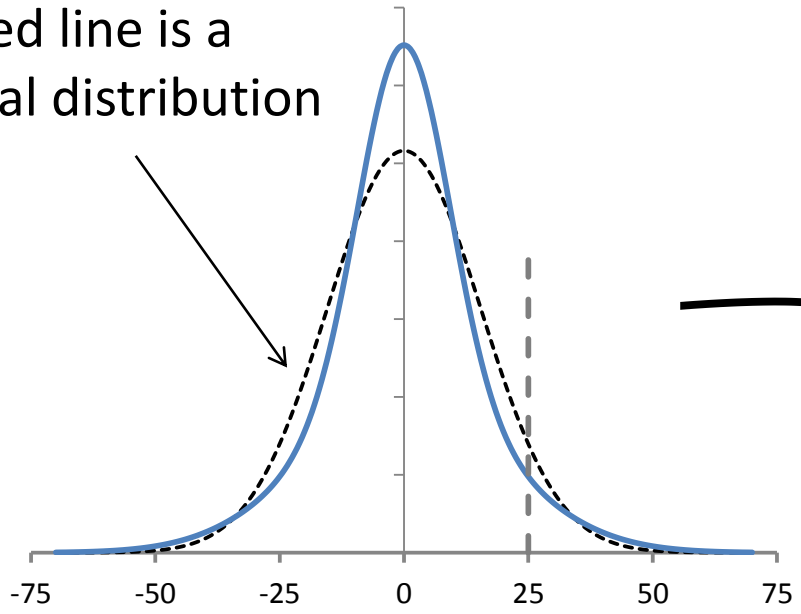
$$\mathrm{VaR}_{0.7}(L) = \inf\{x : P(L \leq x) \geq 0.7\} = 1.5.$$

$$\mathrm{VaR}_{0.65}(L) = \inf\{x : P(L \leq x) \geq 0.65\} = 1.5.$$

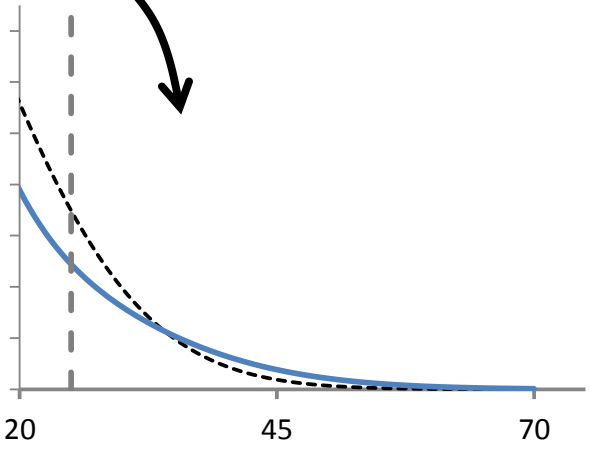
Beyond Value at Risk

- Different behaviours can lead to the same VaR

Dashed line is a normal distribution



Blow this up to see better



Conditional Value at Risk (CVaR)

- One approach is to look at different % values for VaR. The solid line distribution with the fatter tails has a 99% VaR of 41, while the 99% VaR for the normal distribution, is 36.
- An alternative is to use expected shortfall, ("tail value at risk" or "conditional value at risk" (CVaR)).
- CVaR is more natural than VaR. The 95% VaR answers the question "What is the minimum loss amongst the 5% of worst outcomes?". But the 95% CVaR (expected shortfall value) answers the question "What is the average loss amongst the 5% of worst outcomes?"

Mathematical formulation of CVaR when X is continuously distributed

- CVaR can be mathematically formulated as

$$\begin{aligned}
\mathrm{CVaR}_\beta(X) &= \mathbb{E}[X|X > \mathrm{VaR}_\beta(X)] \\
&= \frac{1}{P(X > \mathrm{VaR}_\beta(X))}\mathbb{E}\left[X\mathbb{1}_{X>\mathrm{VaR}_\beta(X)}(X)\right] \\
&= \frac{1}{1-\beta}\mathbb{E}\left[X\mathbb{1}_{X>\mathrm{VaR}_\beta(X)}(X)\right] \qquad (13) \\
&= \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(X)}^{+\infty} x dF_X(dx) \qquad (14)
\end{aligned}$$

  where $\mathbb{1}_A(X) = 1$ for $X \in A$ and 0 otherwise. This is the expected loss conditional on the VaR level being exceeded .

- In the literature of finance, this is also known as Average Value at Risk (AVaR).

Example 10.11

Let $L(\xi) = \xi$ be a loss function where ξ has a density of $d(x) = 0.1e^{-0.1x}$. The loss is in \$1000s. Calculate $\mathrm{CVaR}_{0.95}(L)$.

Solution. First we calculate $\mathrm{VaR}_{0.95}(L)$. By definition

$$\mathrm{VaR}_{0.95}(L) = \mathrm{VaR}_{0.95}(\xi).$$

The property at the tail of the exponential is

$$\int_u^\infty d(x)dx = \int_u^\infty 0.1e^{-0.1x}dx = e^{-0.1u}.$$

Set $e^{-0.1u} = 0.05$. Then we obtain

$$\mathrm{VaR}_{0.95}(\xi) = -10\log_e 0.05 = 29.957.$$

Consequently

$$\mathrm{VaR}_{0.95}(L) = 29.957$$

$$
\begin{aligned}
\mathrm{CVaR}_{0.95}(L) &= \mathrm{CVaR}_{0.95}(\xi) \\
&= \left(\frac{1}{1-0.95}\right)\int_{29.957}^{\infty} x(0.1e^{-0.1x})dx.
\end{aligned}
$$

Note that

$$
\begin{aligned}
\int_{29.957}^{\infty} x(0.1e^{-0.1x})dx &= x(-e^{-0.1x})|_{29.957}^{\infty} + 10\int_{29.957}^{\infty} d(-e^{-0.1x}) \\
&= 29.957e^{-2.9957} + 10e^{-2.9957} \\
&= 1.998.
\end{aligned}
$$

Thus

$$
\mathrm{CVaR}_{0.95}(L) = 1.998/0.05 = 39.96.
$$

References

- M. Allais, "Le comportement de l'homme rationnel devant le risque: critique des postulats et axiomes de l'école Américaine". Econometrica. 21 (4): 503–546.
- Anderson, E. J., Business risk management: models and analysis. John Wiley & Sons, 2013.
- G. Debreu, Continuity Properties of Paretian Utility, *International Economic Review*, 5, 285-293, 1964.
- Hadar, J., W. R. Russell. 1969. Rules for ordering uncertain prospects. Amer. Econom. Rev. 59 25-34.

References

- Hanoch, C., H. Levy. 1969. The efficiency analysis of choices involving risk. Rev. Econom. Stud. 36 335-346.
- C. Puppe, Distorted probabilities and choice under risk, in *Lecture Notes in Economics and Mathematical Systems*, Vol. 363, Springer Science & Business Media 2012.
- J. Von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton University Press, Princeton, 1947.